UNIVERSITY OF LONDON

BSc EXAMINATION 2020

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

CS3003: IT Project Management

Sample Exam Paper Model Answers

Time Allowed: **TWO** hours

Answer ALL questions

Calculators are NOT permitted

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1.

(a) In the context of project Risk Management, indicate whether each of the following statements is true or false. A justification is **not** required.

(i) A Risk Log and a Risk Register are two different terms that refer to the same management product;

(ii) The notion of Risk Appetite is a characteristic of a specific risk;

(iii) All projects have a degree of risk-taking;

(iv) Risk Planning is an activity of the Risk Management process;

(v) The Project Sponsor is accountable for ensuring that all risks are identified, recorded and regularly reviewed;

(vi) It is a valid option to take no action for a known risk.

[12 marks]

Marking Scheme *(partially bookwork, students should demonstrate appropriate understanding of bookwork)*

2 marks per correct answer

Answer

(i) True

(ii) False

(iii) True

(iv) True

(v) False

(vi) True

(b) Explain the five different types of actions that may be applied in order to treat a project risk. [10 marks]

Marking Scheme *(bookwork)*

2 marks per correct and complete answer for each type of action

Answer (from lecture slides, Prince2 actions – any similar answer will get full marks)

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Table 17.1

Prevention	Terminate the risk – by doing things differently and thus removing the risk, where it is feasible to do so. Countermeasures are put in place that either stop the threat or problem from occurring or prevent it having any impact on the project or business
Reduction	Treat the risk – take action to control it in some way where the actions either reduce the likelihood of the risk developing or limit the impact on the project to acceptable levels
Transference	This is a specialist form of risk reduction where the management of the risk is passed to a third party via, for instance, an insurance policy or penalty clause, such that the impact of the risk is no longer an issue for the health of the project. Not all risks can be transferred in this way
Acceptance	Tolerate the risk – perhaps because nothing can be done at a reasonable cost to mitigate it or the likelihood and impact of the risk occurring are at an acceptable level
Contingency	These are actions planned and organised to come into force as and when the risk occurs

(c) You are in the process of planning a project and found that stakeholders have a variety of objectives and requirements. This makes it hard to come up with a plan with which all objectives will be met. Which of the four actions below is probably the most appropriate to take? Briefly **justify** your answer.

- Ask the project stakeholders to build focus groups in order to discuss and reach consensus in conflicting interests;
- Create a comprehensive document describing your point of view and ask stakeholders to review it and sign-off;
- Ask each stakeholder write a statement of work, merge the documents to a project scope statement;
- Use your authority as the project manager to clearly prioritize the different objectives and requirements.

[6 marks]

Marking Scheme *(non-bookwork)*

Full marks for correct answer. No marks for incorrect answer, unless a reasonable explanation is provided.

Answer: The first option, i.e. ask the project stakeholders to build focus groups in order to discuss and reach consensus in conflicting interests, is the best action to take. All other options imply an approach that does not take into account the joint view of stakeholders in a timely manner.

(d) Provide the definition of a project and explain the difference between a project and a program, (i.e. in the sense of a project-related program, not a software program). [6 marks]

Marking Scheme *(bookwork)*

3 marks for definition of project, 3 marks for difference between project and program.

Answer

There are various definitions of a project, e.g. “a temporary endeavour undertaken to create a unique product, service or result. (PMI)” or “a unique, transient endeavour undertaken to achieve a desired outcome” (APM). Any similar definition will suffice as answer.

A Program is a group of related projects to ensure that the strategic objectives of the organisation are met in the most cost effective and efficient manner. This requires the allocation of resources (physical and financial) to all the projects which make up the program. Hence, a program is a collection of, in some way, related projects managed in a coordinated way.

2.

(a) In the context of Configuration Management, provide the definition of each of the following:

(i) Version;

(ii) Codeline;

(iii) Release.

[9 marks]

Marking Scheme *(bookwork)*

3 marks for providing an appropriate definition in each of (i), (ii), (iii). Partial marks for answers the partially explain the concepts at hand (depending on the quality of the answer).

Answer:

(i) Version: An instance of a configuration item that differs, in some way, from other instances of that item. Versions always have a unique identifier, which is often composed of the configuration item name plus a version number.

(ii) Codeline: is a set of versions of a software component and other configuration items on which that component depends.

(iv) Release: A release is a complete and consistent set of products, which forms a fixed reference point in the development of the end outcome, e.g. a system version ready to become available to end-users.

(b) A company is about to buy a Relational Database Management System (RDBMS). The system consists of four components, namely A1, A2, A3 and A4, which are licensed independently, according to the following scheme:

- A1 is licensed on a per user basis, costing £200 per user that will have access to the system.
- A2 is licensed on a per user basis, costing £100 per user that will have access to the system.
- Components A3 and A4 are licensed based on the number of CPU's (processors) of the hardware that the RDBMS will be installed on. The license cost of A3 is £10,000 per installed CPU and the license cost of A4 is £20,000 per installed CPU.

It is estimated that in order to adequately perform in the production environment, the RDBMS requires 2 CPUs for up to 200 users, 4 CPUs for 201 to 500 users and 8 CPUs for 501 to 1000 users. Also, the new system will have a disaster recovery site of a processing capacity that is equal to the production one. The RDBMS license fee for the disaster recovery site is zero.

Additionally, starting from the second year, the company is obliged to pay the vendor of the RDBMS an annual maintenance fee of 20% over the total license fee.

Finally, each CPU of the production and disaster environments is acquired at a cost of £5,000 and has an annual maintenance fee of 10% over the acquisition cost. The CPU maintenance fee also starts from the second year.

If variable N denotes the number of users and variable M the number of CPUs, based on the previous facts, perform the following tasks:

i. Write down the formulas to calculate the 5-year Total Cost of Ownership (TCO) of the investment that the company has to make for the RDBMS. [10 marks]

ii. Calculate the 5-year TCO of the RDBMS for 600 users. [6 marks]

Notes: All required CPUs will be acquired from the market, i.e. the system will not be installed on an existing hardware (H/W) infrastructure. The time-value of money should not be taken into account in the TCO calculation.

Marking Scheme *(non-bookwork)*

4 marks for correct licensing calculation, 4 marks for S/W annual maintenance, 2 marks for H/W acquisition and maintenance, 6 marks for TCO calculation for 600 users.

Answer

If N denotes the number of users and M the number of CPUs then the 5-year TCO is given by the following formulae:

TCO = (SW Acquisition Cost + SW Maintenance Cost + H/W Acquisition Cost (of both Production and Disaster) + H/W Total Maintenance Cost) =

(Cost $_{\text{S/W licensing}}$ + Cost s/w $_{\text{Maintenance (5-years)}}$ + Cost $_{\text{CPUs (Prod, Disaster)}}$ +Cost H/W $_{\text{Maintenance (5-years)}}$)

- Cost $_{\text{S/W licensing}}$ = 200*N + 100*N + 30,000*M with M=2 (0<N<=200), M=4 (200<N<=500), M=8 (500<N<=1000). No S/W License fee for the Disaster Site.
- Cost $_{\text{S/W Maintenance (5-years)}}$ = 4*[20% (Cost $_{\text{S/W licensing}}$)]
- Cost $_{\text{CPUs}}$ = 5,000*M (Production) + 5,000*M (Disaster) with M=2 (0<N<=200), M=4 (200<N<=500), M=8 (500<N<=1000).
- Cost H/W $_{\text{Maintenance (5-years)}}$ = 4*[10% (Cost $_{\text{CPUs}}$)]

If N = 600 then M = 8 and

- Cost $_{\text{S/W licensing}}$ = 200*600 + 100*600 + 30,000*8 = 120,000+60,000+240,000 = £420,000
- Cost $_{\text{S/W Maintenance (5-years)}}$ = 4*[20% (420,000)] = 4*84,000 = £336,000
- Cost$_{\text{CPUs}}$ = 10,000*8 = £80,000
- Cost H/W $_{\text{Maintenance (5-years)}}$ = 4*[10% (80,000)] = £32,000

Hence, 5-year TCO for 600 Users = 420,000+336,000 + 80,000 + 32,000 = £868,000

(c) Outline four methods often used in requirements elicitation from stakeholders. [8 marks]

Marking Scheme *(bookwork)*

2 marks for each method.

Answer (from lecture slides)

Methods to elicit requirements from stakeholders: Interviews, Brainstorming workshops, Questionnaires and Surveys, Use Case Analysis and User Stories, Ethnography (observation of current working practices), Prototypes and Modelling, Demonstrations of current operating environment, 5-Why Analysis, Fishbone diagrams, Use experience/ examples / blueprints from previous projects.

3.

(a) The activities of a project and respective durations and dependencies are shown in the following table.

Activity	Duration (in days)	Predecessors
A	20	
B	30	A
C	40	A
D	33	A
E	36	B
F	10	E, D
G	4	C, F
H	2	G

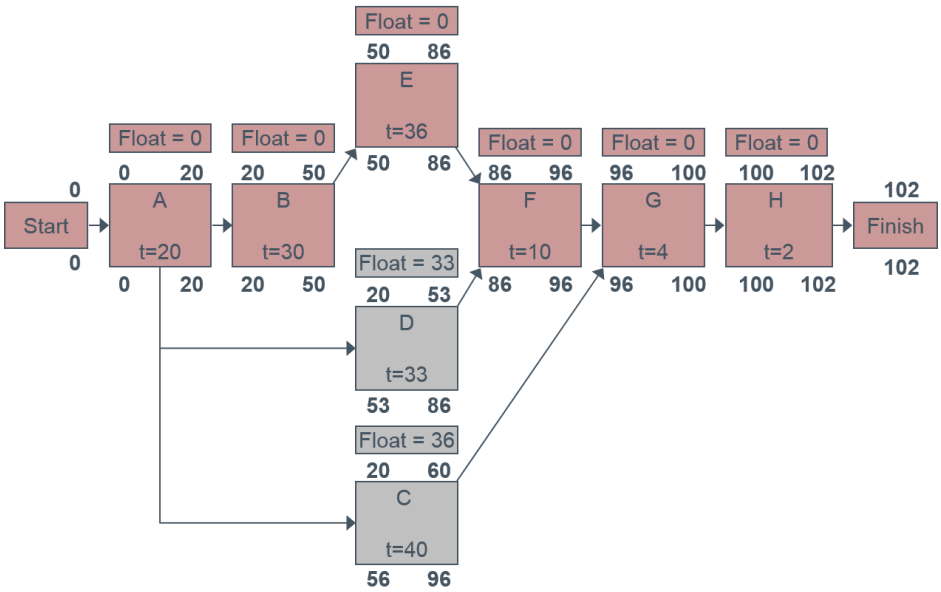
(i) Draw a network diagram for this project including all floats; [10 marks]

(ii) Mark the project's critical path on the diagram; [3 marks]

Marking Scheme *(non-bookwork)*

2 marks for each correct node/sequence/float in the diagram, excluding node H. Deduction of 2 marks for each incorrect node/float. Full marks for correct critical path

Answer (critical path is shown in purple)



(b) You have been assigned as the project manager for a new IT project in your organization. The project deals with the implementation of a new accounting system. You are about to present the project at the "kick-off" meeting arranged for that purpose. The meeting participants will mainly be the project sponsor and various stakeholders, i.e. a management level audience.
Provide a proposal of an outline of the presentation you would make in such a meeting. The proposal should consist of 5 slides you believe are most suitable for the audience and goal of the meeting. The sequence of your slides should also make sense. For each slide explain the rationale for your choice of content and its sequence. Note that you do not have to produce an actual presentation, but just an outline in which, for each slide, you are expected to provide a suitable slide title and a brief description of content.

[10 marks]

Marking Scheme *(non-bookwork)*

2 marks for each slide that makes sense, appropriately explained (including rationale)

Answer

Slide 1. Project high-level scope, concrete objectives and success criteria

Slide 2. Project governance structure and concrete project teams

Slide 3. Project high-level plan/key milestones/dates

Slide 4. Project budget

Slide 5. Project risks (initial set)

Note. Possible slides also include next steps, key decisions to be made by sponsor and stakeholders. Sequence may also be different if rationale is explained. A slide with description of the system will not be accepted as a correct answer (since the presentation is on the project, not on the system at hand)

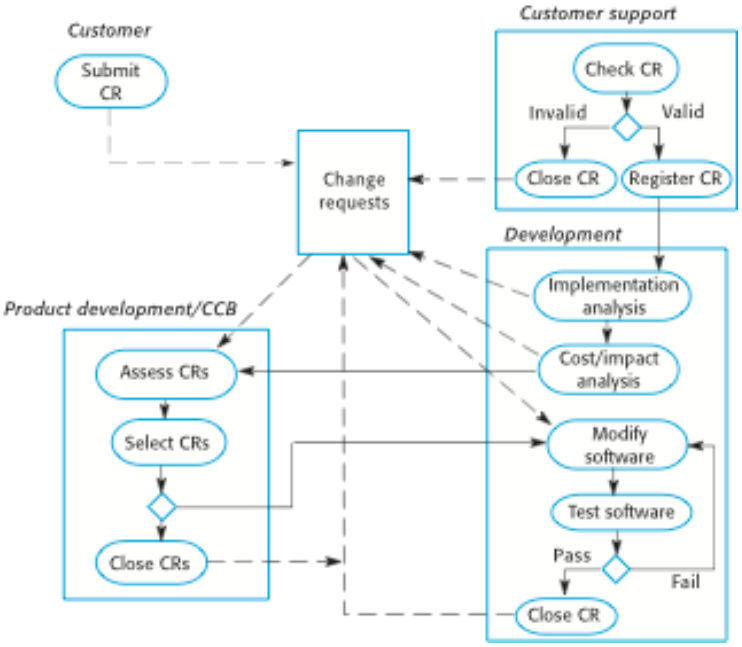
(d) Draw a diagram depicting the Change Management process of a software product development IT project and outline the activities involved.

[10 marks]

Marking Scheme *(bookwork)*

1 mark for each item on the diagram, including a reasonable description.

Answer (from lecture slides, full answer should include a description/outline of the activities involved)


Customer
Submit CR
Customer support
Check CR
Invalid
Valid
Close CR
Register CR
Change requests
Development
Implementation analysis
Cost/impact analysis
Modify software
Test software
Pass
Fail
Close CR
Product development/CCB
Assess CRs
Select CRs
Close CRs

END

GK